



BGFlow: Brightness-guided normalizing flow for low-light image enhancement[☆]

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ABSTRACT

Low-light image enhancement poses significant challenges due to its ill-posed nature. Recently, deep learning-based methods have attempted to establish a unified mapping relationship between normal-light images and their low-light versions but frequently struggle to capture the intricate variations in brightness conditions. As a result, these methods often suffer from overexposure, underexposure, amplified noise, and distorted colors. To tackle these issues, we propose a brightness-guided normalizing flow framework, dubbed BGFlow, for low-light image enhancement. Specifically, we recognize that low-frequency sub-bands in the wavelet domain carry significant brightness information. To effectively capture the intricate variations in brightness within an image, we design a transformer-based multi-scale wavelet-domain encoder to extract brightness information from the multi-scale features of the low-frequency sub-bands. The extracted brightness feature maps, at different scales, are then injected into the brightness-guided affine coupling layer to guide the training of the conditional normalizing flow module. Extensive experimental evaluations demonstrate the superiority of BGFlow over existing deep learning-based approaches in both qualitative and quantitative assessments. Moreover, we also showcase the exceptional performance of BGFlow on the underwater image enhancement task.

1. Introduction

Images captured under low-light conditions often suffer from severe degradation due to low contrast, intensive noises, and invisible details. The degraded low-light images are frustrating for some downstream visual tasks, such as object and action recognition at night [1]. In the past few years, many scholars have devoted themselves to studying low-light image enhancement, which has become an important area of low-light image vision applications.

Currently, various deep learning-based approaches have significantly progressed in low-light image enhancement tasks [2–5]. It can be roughly split into two groups: the retinex-based and non-retinex-based methods. The retinex-based methods combine deep learning with retinex theory in traditional models to further improve the generalization ability of the model. Chen et al. [6] separated the input images into their corresponding reflectance and illumination components by utilizing an encoder–decoder network to further modify the illumination component. However, the enhanced results suffered from detail distortion since the loss function cannot effectively constrain the intermediate image. Furthermore, some methods [7–9] followed the

RetinexNet [6] and considered the impact of noise during the enhancement process. The common feature of these methods is the introduction of additional training constraint losses. Nevertheless, the result of retinex-based methods [3,10] inevitably undergoes color distortion and noise amplification since there are no real reflections and illuminations in practice. Non-retinex-based methods often focus on the design of the network structure and consider introducing additional information for low-light image enhancement. Recently, the wavelet transform has proven to be an effective technique in deep learning-based methods for low-light image enhancement. For instance, HWMNet [11] innovatively introduced half wavelet attention blocks, which integrate wavelet transform with attention mechanisms. This integration adaptively captures a rich and diverse set of features, thereby significantly improving the enhancement effect. DiffLL [12] expanded the utility of wavelet transform through an innovative fusion with diffusion models. This approach accelerates the inference process, reduces computational resources, and ensures the preservation of information integrity. However, the full potential offered by the wavelet transform still needs to be further explored and utilized.

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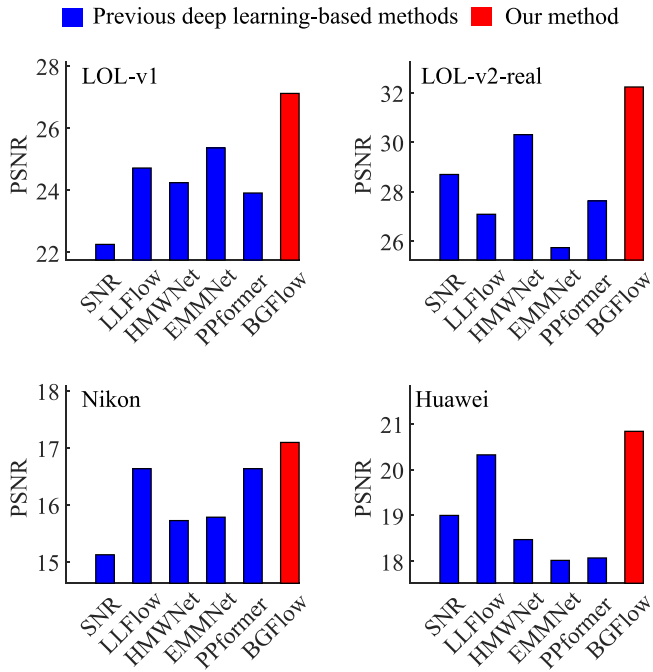


Fig. 1. Peak signal-to-noise ratio (PSNR) on LOL-v1, LOL-v2-real and LSRW dataset which contains the Nikon and Huawei datasets. Using the same network design, our method consistently outperforms the state-of-the-art methods across four distinct benchmarks in terms of PSNR. Each individual plot compares results on a single benchmark dataset.

Moreover, the conditional normalizing flow model [13–15] is a powerful generative model that introduces conditional variables into the standard normalizing flow framework, enabling more fine-grained control over data distribution. The role of the condition is crucial as it allows the model to generate data samples with specific attributes based on the given condition. WaterFlow [14] incorporated heuristic priors by integrating the characteristics of underwater light absorption and scattering, proposing a conditional normalizing flow model for underwater image enhancement. By introducing semantic features as conditional guidance, the model not only improves the visual quality of the images but also significantly boosts performance in object detection tasks. HCFlow [15] used low-resolution images as conditional inputs to learn a bijective mapping between high-resolution and low-resolution image pairs. Specifically, the low-resolution image guides the generation process of the high-resolution image. Therefore, one of the challenges of conditional normalizing flow models is effectively incorporating conditional information into the flow transformations while maintaining computational efficiency and generation quality. However, extracting effective conditions for conditional flow models poses a significant challenge.

In order to deeply explore the potential of the wavelet transform for application in low-light image enhancement tasks and efficiently extract conditional guidance to the normalizing flow model, we propose a brightness-guided normalizing flow model for low-light image enhancement, named BGFlow. BGFlow can effectively address the issues of overexposure, underexposure, and color distortion prevalent in current low-light image enhancement methods. It reconstructs brightness and color information while maintaining realistic and natural-looking results. BGFlow inherits the advantages of the normalizing flow model, particularly its invertibility, and cleverly exploits the characteristics of low-light image in the wavelet domain. By integrating the brightness feature from the brightness feature extraction block (BFEB) into the normalizing flow model, the proposed method can effectively restore both image brightness and color information. Specifically, BFEB focuses on recovering the low-frequency sub-bands of the low-light input in

the wavelet domain, which contain a significant amount of brightness information. The encoding process of BFEB generates brightness feature representation at different scales, playing a crucial role in guiding the normalizing flow model to obtain the latent representation and ultimately produce satisfying visual quality. In BGFlow, the forward process of the normalizing flow model can be viewed as an image degradation process, where the transition from the normal image to the latent representation of the degradation of image detail information. Due to the invertible nature of the normalizing flow model, it can reconstruct the image losslessly during the inference phase. Additionally, unlike methods [16,17] that reconstruct images directly from Gaussian noise, our approach reconstructs the image from a latent representation containing brightness information. As shown in Fig. 1, BGFlow exhibits superior performance compared to the state-of-the-art approaches across four commonly used datasets.

The main contributions are summarized as follows:

- (1) We come up with a novel brightness-guided normalizing framework (BGFlow) for dealing with the low-light image enhancement task, wherein the brightness information of images in the wavelet domain is extracted to assist reversible networks in fitting complex conditional distributions by a designed brightness-guided affine coupling layer.
- (2) We propose a novel wavelet-domain encoder that provides rich brightness information guidance to normalizing flow modules. The transformer block, which serves as the core component of the wavelet-domain encoder, is responsible for extracting brightness information across various scales in the wavelet domain. By injecting brightness features at various scales into the normalizing flow model, the BGFlow is able to reconstruct more precise and realistic output during the inference process.
- (3) We showcase the effectiveness of the proposed BGFlow through a series of extensive experiments on seven datasets. BGFlow can achieve significant performance advantages on the underwater image enhancement and semantic segmentation tasks.

The rest of the paper is organized as follows. Section 2 reports the related works, which include traditional low-light image enhancement methods, deep learning-based low-light image enhancement methods, and the normalizing flow model. Section 3 describes the proposed BGFlow framework and the associated loss functions. Section 4 presents the implementation details, introduces the result analysis, and presents an ablation study. Finally, Section 5 concludes our method.

2. Related works

In this section, we present related works on low-light image enhancement and normalizing flow models. Among them, low-light image enhancement methods can be divided into traditional based methods and deep learning-based methods.

2.1. Low-light image enhancement

2.1.1. The traditional methods

For the traditional low-light image enhancement methods, the most direct way is to amplify the brightness of the image. Histogram equalization-based methods [18,19] can markedly boost image contrast and brightness. Pizer et al. [19] addressed the slow speed of adaptive histogram equalization and the excessive enhancement of noise generated in relatively homogeneous regions by emphasizing the pixel contribution to the histogram and cropping the adaptive histogram equalization. Abdullah et al. [20] proposed a dynamic histogram equalization algorithm based on traditional histogram equalization which is proposed to achieve the enhancement effect without missing any details. However, these algorithms are inferior in reducing or suppressing noise, and may cause color distortion or exacerbate noise. He et al. [21] introduced an advanced post-processing method that

effectively enhances low-light images by optimizing gamma correction maps. Moreover, some methods [22,23] based on retinex theory attempted to estimate the illumination and the reflectance in the image domain. Guo et al. [22] introduced a direct and efficient approach for enhancing low-light images, wherein the initial brightness is determined by selecting the maximum value between the R, G and B channels. The enhanced illumination was obtained through iterative updates using the alternating direction minimization technique. Besides, considering the influence of noise, a robust retinex model [24] was proposed to further improve method performance.

2.1.2. The deep learning-based methods

With the rapid development of extensive datasets and advancements in computational capabilities, learning-based methods have shown great potential in low-light image enhancement. Lv et al. [25] proposed a multi-branch low-light image enhancement model to address the multi-functional requirements that single-branch or simple neural networks could not meet, such as simultaneous brightness and contrast enhancement, artifact removal, and noise reduction. Similarly, Lu et al. [26] developed a two-branch exposure fusion network for enhancing low-light images, where the network processes images with different exposure levels to achieve better enhancement. Fan et al. [11] further improved low-light image enhancement methods by integrating wavelet-based attention mechanisms into existing network architectures, specifically M-Net+. This integration improved the network's capability to focus on important features, thereby improving the quality of the enhanced images. Wang et al. [27] proposed a method that calculates the global illumination estimate of a low-light image and adjusts the illumination accordingly. This approach, combined with connections to the original input to supplement details, effectively enhances the low-light image. Moreover, Zheng et al. [28] introduced learnable prior information and dynamically adjusted these priors during the training process, thereby better adapting them to enhance specific images. This adaptive approach ensures that the enhancement process is more tailored to the unique characteristics of each image.

Various approaches have combined Retinex theory with convolutional neural networks to enhance low-light images effectively [6,7]. To address the issue that increasing the brightness of dark areas often amplifies hidden degradations, Zhang et al. [7] decomposed the image into two components: illumination, responsible for light regulation, and reflectivity, responsible for degradation removal. This decoupling of the original image space into two smaller subspaces aims to achieve better regularization and learning effects. KinD_plus [8] was an improved version of KinD [7] that builds upon this concept. Wu et al. [29] introduced a deep unfolding network based on Retinex theory, called URetinex, for low-light image enhancement. By unfolding the Retinex model through deep learning methods, URetinex combines the theoretical advantages of traditional methods with the powerful feature extraction capabilities of deep learning to achieve efficient low-light image enhancement.

Although these methods can achieve significant improvements in image quality, they inevitably lead to issues such as loss of details and blurring. To address these challenges, some approaches have exploited attention mechanisms to acquire more reliable mappings [3,30]. Furthermore, traditional low-light image enhancement methods often amplify noise in the image. To overcome this problem, Xu et al. [3] introduced a signal-to-noise ratio-aware mechanism that suppresses noise while enhancing the image, thereby improving overall quality. Dang et al. [31] addressed the high computational cost associated with self-attention mechanisms in low-light image enhancement tasks by leveraging pixel-level and patch-level cross-attention mechanisms. This approach significantly enhances color consistency, brightness, and contrast in images. Additionally, Cai et al. [32] proposed a transformer framework that utilizes illumination guidance to enhance low-light images. This method employs illumination representations to facilitate

the modeling of non-local interactions among regions with distinct brightness conditions.

Despite significant advancements in deep learning techniques, there remains ample room for improvement. Ye et al. [33] introduced a novel approach to enhancing low-light images by leveraging external memory. This innovation allows test samples to dynamically adjust by aligning more closely with the diverse distribution of reference images in the dataset. Moreover, integrating transformers into the core network enables effective capture of spatially redundant information over long distances, thereby enhancing overall model performance. Although attention-based methods have demonstrated effectiveness, they still exhibit varying degrees of artifacts and detail loss in their results. Addressing these challenges, Jiang et al. [12] proposed a low-light image enhancement method based on a wavelet conditional diffusion model. This approach utilizes wavelet transforms to expedite inference and reduce computational demands. Furthermore, incorporating a high-frequency recovery module ensures meticulous restoration of image details, promising more logical refinements to the enhancement process.

2.2. Normalizing flow

As a typical generative model of natural images, normalizing flow-based methods [17,34] formulate a complex probability distribution $p_y(y|\theta)$ by a sequence of invertible and differentiable neural networks f_θ . Conditional normalizing flow [35] inherited the advantages of normalizing flow, and it is easy to obtain the inverse of the generated samples and likelihood. For the task of low-light image enhancement, f_θ encodes the paired image samples (x, y) to a latent representation $z = f_\theta(y; x)$, where z is often assumed to be a simple Gaussian distribution $p_z(z)$ and f_θ consists of a set of invertible layers $h^1, h^2, \dots, h^N$, where $h^{n+1} = f_\theta^n(h^n, g_\phi(x))$, $n \in (0, N-1)$, $h^0 = y$ and $h^N = z$. $g_\phi(x)$ can be a standard neural network that encodes x to the latent space and extracts efficient condition features for all flow layers. According to the change of variable theorem, the conditional distribution $p_{y|x}(y|x, \theta)$ can be expressed as follows:

$$p_{y|x}(y | x, \theta) = p_z(f_\theta(y; x)) | \det \frac{\partial f_\theta}{\partial y}(y; x) |. \quad (1)$$

Then the negative log-likelihood is used to optimize the Eq. (1). This can be expressed as :

$$\mathcal{L}(\theta, \phi; x, y) = -\log p_z(z) - \sum_{n=0}^{N-1} \log \left| \det \frac{\partial f_\theta^n}{\partial h^n}(h^n, g_\phi^n(x)) \right|. \quad (2)$$

In Eq. (2), it only need to estimate the logarithmic-determinant of the Jacobian $\frac{\partial f_\theta^n}{\partial h^n}$ for each flow layer by leveraging standard SGD-based optimization manner.

Recently, normalizing flow-based approaches [36–38] have been used in the conditional generation of images. Abdelhamed et al. [39] proposed a noise modeling method based on conditional normalizing flows, which has been used to generate and adjust noise distribution. Lugmayr et al. [16] introduced a super resolution high-quality image generation method based on regularized flow. Wang et al. [40] obtained great success via employing conditional image generation in low-light image enhancement. Shi et al. [38] proposed a novel network guided by cartoon and texture for enhancing low-light images.

3. Method

For low-light image enhancement, BGFlow learns the mapping from the low-light image x_{low} to the enhanced image $\hat{y}$. The overall framework includes a brightness feature extraction block (BFEB) and a brightness-guided normalizing flow module (BGNF) which comprises brightness invertible blocks (BIBs). First, the BFEB extracts an intermediate image and the brightness features at different scales. These features are crucial for constructing complex conditional guidance

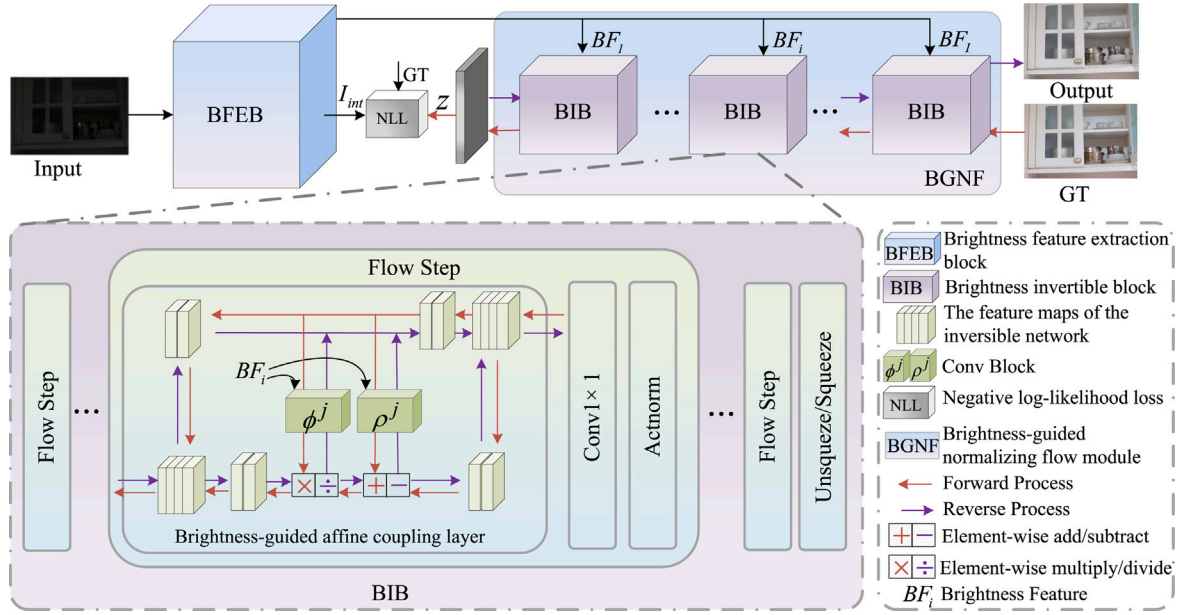


Fig. 2. The overall framework of the proposed BGFlow. BGFlow consists of two modules: the brightness feature extraction block (BFEB) and the brightness-guided normalizing flow module (BGNF). BFEB captures multi-scale brightness information in the wavelet domain. BGNF involves brightness invertible blocks (BIBs) which are designed based on a conditional normalizing flow model. BGNF can couple the brightness features from different scales with the features of the normalizing flow model.

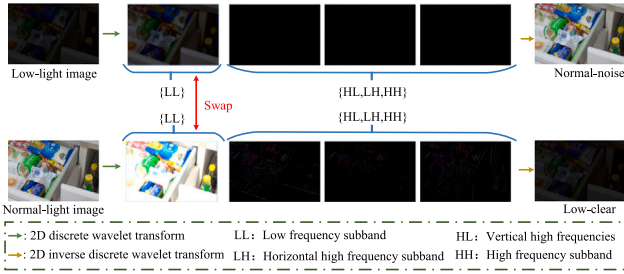


Fig. 3. The characteristics of real low-light images in the wavelet domain. We notice that wavelet transformation can separate brightness and color information. By analyzing the different sub-bands obtained from the wavelet transform, we have found that exchanging the low frequency sub-bands can significantly enhance the brightness and color of low-light images.

which can be used to guide the conditional normalizing flow to reconstruct high-quality images. Second, the brightness-guided affine layer couples the features from BFEB with those in BIBs. Finally, BGNF simulates the output of BFEB during the forward process and obtains a high-quality enhanced image during the reverse process. Fig. 2 illustrates the complete pipeline of the proposed BGFlow.

3.1. Brightness feature extraction block

Wavelet transform (WT) is a mathematical tool widely utilized in signal or image processing for decomposing signals or images into their frequency components at various scales. This transform boasts the capability of multi-scale analysis, breaking down signals or images into frequency components across different scales, thereby offering an insightful analysis of the local frequency information within the signal or image. Another noteworthy feature of WT is its invertibility, meaning that the original signal or image can be reconstructed based on the wavelet coefficients.

In this paper, we delve into the characteristics of real low-light images in the wavelet domain, as depicted in Fig. 3. Specifically, swapping the low-frequency sub-bands of a noisy low-light image with those of a clear normal-light image results in a noisy normal-light image and a

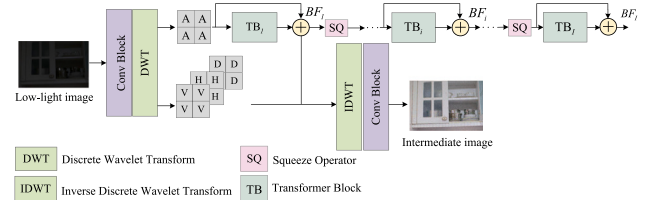


Fig. 4. The architecture of the brightness feature extraction block (BFEB).

clear low-light image. It suggests that low-frequency subbands retain the overall structure and visually critical information of the image. In addition, noise is usually present in the high-frequency subbands, while there is less noise in the low-frequency subbands. Focusing on low-frequency subbands effectively attenuates noise interference and enhances the brightness of the image. This inspires us that we can utilize WT to improve the visual quality of low-light images.

Furthermore, conditional normalizing flow models have shown advantages in image generation and reconstruction tasks, but how to construct task-specific conditions remains to be explored. To address this challenge, we develop a novel model named BFEB, designed to capture brightness information within the low-frequency subbands. Subsequently, the brightness features are injected into BGNF to strengthen the model's capabilities.

Given an low-light input x_{low} , BFEB aims to extract brightness features at different scales and an intermediate image I_{int} . Fig. 4 depicts the structure of BFEB. First, we transform x_{low} from the image space to the feature space through a convolutional block. Subsequently, WT is employed to decompose the feature space into two parts: brightness and detail space. In this process, we mainly focus on extracting reasonable brightness information. To accurately extract brightness information, we adopted the transformer structure [41], which is characterized by its high computational efficiency and excellent performance, as the main network component of BFEB. In order to construct more complex feature representations, we use the method [42] to obtain brightness

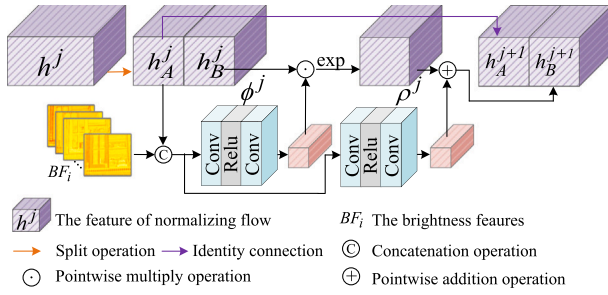


Fig. 5. The architecture of the brightness-guided affine coupling layer.

features at different scales. The entire process can be formulated as follows:

$$\begin{aligned}
 AA, VV, HH, DD &= \text{DWT}(\text{CB}_1(x_{low})), \\
 BF_1 &= \text{TB}_1(AA) + AA, \\
 BF_i &= \text{TB}_i(\text{SQ}(BF_{i-1})) + (\text{SQ}(BF_{i-1})), i \in \{2, \dots, I\},
 \end{aligned} \quad (3)$$

where $\text{TB}_i(\cdot)$ is transformer block proposed in the method [41] and $i \in \{1, \dots, I\}$ indicates different resolution. $BF_i, i \in \{1, \dots, I\}$ corresponds to different resolution of brightness variables extracted by $\text{TB}_i(\cdot)$ with a skip connection. $\text{CB}_1(\cdot)$ is a procedure of shallow feature extraction. It consists of two convolutional layers, the first of which is followed by a ReLU activation function. $\text{SQ}(\cdot)$ is the downsampling operation which maps features from $C \times H \times W$ to $C \times H/2 \times W/2$ via bilinear interpolation. We use the Haar wavelet transform [43] in BFEB and its filters can be defined as follows:

$$\begin{aligned}
 f_{AA} &= \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, f_{VV} = \begin{bmatrix} -1 & -1 \\ 1 & 1 \end{bmatrix}, \\
 f_{HH} &= \begin{bmatrix} -1 & 1 \\ -1 & 1 \end{bmatrix}, f_{DD} = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix},
 \end{aligned} \quad (4)$$

where f_{AA} is low-pass filter, and f_{VV}, f_{HH}, f_{DD} are detail filters. They are employed to build four sub-bands AA, VV, HH, DD with the size of $H/2 \times W/2$ by decomposing the input with the size of $H \times W$.

To ensure that the BFEB can extract rich brightness features, we remap the brightness feature BF_1 to the image space to generate the intermediate image I_{int} . I_{int} will be used to constrain the latent representation z , which is the product of the forward degradation process of BGNF. In detail, a convolution layer is used to transform the output of the inverse discrete wavelet transform (IDWT) to image space. The entire process can be formulated as follows:

$$I_{int} = \text{CB}_2(\text{IDWT}([BF_1, VV, HH, DD])), \quad (5)$$

where the structure of CB_2 is similar to the CB_1 , except that the last layer uses a sigmoid activation function.

3.2. Brightness-guided normalizing flow module

The BGNF module contains three BIBs which are based on the Glow architecture [17]. BIB contains four components: a squeeze layer, an invertible 1×1 convolution layer, a brightness-guided affine coupling layer, and an actnorm layer. The details of the components are as follows:

Squeeze layer. In order to extract rich features, the squeeze layer [17] is used to compress the resolution and map features from $C \times H \times W$ to $4C \times H/2 \times W/2$.

Invertible 1×1 convolution. This operation [17] is similar to a regular convolution operation, and the size of the convolution kernel needs to be restricted to 1.

Brightness-guided affine coupling layer. A brightness-guided affine coupling layer leverages brightness information to strengthen the capability of the normalizing flow model in capturing intricate

conditional distributions. As depicted in Fig. 5, the features h^j from the normalizing flow model are evenly split along the channel dimension into two parts, h_A^j and h_B^j . Here, h_A^j is concatenated with the brightness feature BF_i and subsequently processed through two distinct feature extraction networks, ϕ^j and ρ^j , designed to extract scale and bias parameters, respectively. These parameters are then incorporated into the normalizing flow model via an affine transformation. Ultimately, the resulting feature h_B^{j+1} is concatenated with feature h_A^{j+1} along the channel dimension to generate a coupled feature that preserves the original input size. The brightness-guided affine coupling operation can be succinctly expressed as follows:

$$\begin{aligned}
 h_A^{j+1} &= h_A^j, \\
 h_B^{j+1} &= \exp\left(\phi^j\left([h_A^j, BF_i]\right)\right) \cdot h_B^j + \rho^j\left([h_A^j, BF_i]\right),
 \end{aligned} \quad (6)$$

where BF_i represents the conditional brightness feature extracted by the BFEB. The functions $\phi^j(\cdot)$ and $\rho^j(\cdot)$ consist of two separate networks, each comprising convolutional layers followed by ReLU activation functions. These networks are meticulously crafted to estimate the scale and bias parameters, respectively. This approach effectively tailors the conditional brightness feature BF_i to suit specific requirements, enhancing the precision and adaptability of the overall model.

Actnorm. Actnorm [17] normalizes each channel and sets the mean value of each channel to 0 and the variance to 1. Additionally, it adjusts the mean value and variance of the normalizing activation function by performing an affine transformation.

3.3. Loss function

On the one hand, the intermediate image I_{int} should have adequate brightness details, enabling the BFEB to effectively provide brightness guidance to the BGNF. Ensuring that I_{int} is consistent with the reference image y_{ref} guarantees sufficient brightness information in I_{int} . On the other hand, aligning I_{int} with the latent representation z is crucial to realize the information coupling between the BFEB and the BGNF. Achieving a good balance between these aspects is key to optimizing network performance. We adopt a specific training approach [40] for the overall framework, encompassing both the encoder and the reversible network. The objective function to constrain BGFlow as follows:

$$\begin{aligned}
 \mathcal{L}(\theta, \phi; x_{low}, y_{ref}) &= -\log \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(z - r(y_{ref}, I_{int}))^2}{2}\right) \\
 &\quad - \sum_{n=0}^{N-1} \log \left| \det \frac{\partial f_{\theta}^n}{\partial h^n}(h^n, g_{\phi}^n(x_{low})) \right|,
 \end{aligned} \quad (7)$$

where $r(a, b)$ is defined as a random selection function, and its definition is as follows:

$$r(a, b) = \begin{cases} a & \alpha \leq p, \\ b & \alpha > p, \end{cases} \quad \alpha \sim U(0, 1), \quad (8)$$

where p is the threshold parameter.

Considering that Eq. (7) only constrains the forward process of the conditional normalizing flow model, we attempt to further refine the network results. We fine-tune the reverse process of the normalizing flow model using the $\mathcal{L}_1$ norm and structural similarity (SSIM) loss function [44]. The output of the model show that our fine-tuning training further boosts the performance of the model. The loss function is expressed as follows:

$$\mathcal{L} = \|\hat{y} - y_{ref}\|_1 + \lambda(1 - \text{SSIM}(\hat{y}, y_{ref})), \quad (9)$$

where $\hat{y} = f_{\theta}^{-1}(I_{int}, g_{\phi}(x_{low}))$ is the output of BGFlow. The value of λ is set to 0.01 in order to achieve a balanced weighting of various losses.

4. Experiments

This section conducts experiments, including datasets, implementation details, result analysis, and ablation study, to evaluate the performance of BGFlow. In addition, our BGFlow can also be applied to underwater image enhancement and semantic segmentation task.

Table 1

Quantitative comparison on the LOL-v1, LOL-v2-real, Nikon and Huawei datasets. $\uparrow$ ($\downarrow$) indicates that larger (smaller) values correspond to better quality. The best result is highlighted in red, while the second-best result is highlighted in blue.

Methods	LOL-v1			LOL-v2-real			Nikon			Huawei		
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
MBLLEN [25]	17.56	0.729	0.174	17.30	0.681	0.222	15.39	0.397	0.257	15.53	0.402	0.444
URetinetex [29]	21.33	0.833	0.121	21.22	0.859	0.099	16.53	0.452	0.214	19.14	0.557	0.361
SNR [3]	22.26	0.827	0.140	28.69	0.876	0.157	15.12	0.465	0.257	19.00	0.581	0.416
LLFlow [40]	24.71	0.859	0.137	27.08	0.901	0.100	16.64	0.452	0.215	20.32	0.561	0.360
HWMNet [11]	24.04	0.849	0.115	29.95	0.908	0.082	15.72	0.470	0.237	18.47	0.579	0.386
CUE [28]	21.67	0.769	0.224	22.56	0.799	0.210	16.39	0.464	0.249	18.75	0.550	0.397
Retinexformer [32]	25.05	0.843	0.131	28.78	0.871	0.113	15.86	0.440	0.212	19.19	0.554	0.320
DiffLL [12]	26.14	0.840	0.118	28.75	0.871	0.102	15.95	0.359	0.401	18.05	0.407	0.482
EMNet [33]	24.54	0.861	0.085	30.00	0.917	0.049	15.78	0.451	0.202	18.02	0.536	0.325
PPformer [31]	23.91	0.841	0.120	27.62	0.882	0.097	16.63	0.471	0.210	18.07	0.554	0.378
BGFlow (Ours)	27.11	0.873	0.110	32.23	0.920	0.071	17.09	0.510	0.250	20.83	0.537	0.364

4.1. Datasets

We evaluate BGFlow on reference and non-reference datasets. Reference datasets include LOL-v1 dataset [6], LOL-v2-real dataset [45], and LSRW dataset [46]. Non-reference datasets contains LIME [22], VV,¹ DICM [47], and MEF [48]. LOL-v1 dataset contains 500 pairs of images. Among them, 485 pairs are the training set and 15 pairs are the test set. LOL-v2-real consists of 689 image pairs, while the test set contains 100 image pairs. LSRW dataset includes 5650 image pairs, with 3170 pairs captured using a Nikon D7500 camera and 2480 pairs captured using Huawei P40 Pro. A total of 20 pairs from the Nikon D7500 camera and 30 pairs from the Huawei P40 Pro are exclusively utilized for testing purposes. LIME, MFE, DICM and VV are several widely used reference-free datasets for evaluating the performance of low-light image enhancement methods, and they contain 10, 17, 69 and 24 low-light images from different scenes, respectively.

4.2. Implementation details

We conduct the center crop of size 160×160 and use the Adam optimizer to train our BGFlow. In the training step, BGFlow is trained for 3×10^4 iterations on LOL-v1 datasets. The initial learning rate is set to 5×10^{-4} and it is decreased with a factor of 0.5 at 1.5×10^4 , 2.25×10^4 , 2.7×10^4 , 2.85×10^4 iterations, respectively. Our proposed BGFlow is trained on a single NVIDIA RTX 3090Ti GPU with the PyTorch framework.

4.3. Result analysis

We qualitatively and quantitatively compare the results of the proposed BGFlow with ten state-of-the-art image enhancement methods: MBLLEN [25], URetinetex [29], SNR [3], LLFlow [40], HWMNet [11], CUE [28], Retinexformer [32], DiffLL [12], EMNet [33], PPformer [31]. To ensure a comparison that is both fair and rigorous, we utilize the source code provided by the respective authors and adhere strictly to identical experimental conditions across all assessments. First, we train the model using the LOL-v1 dataset. Then, we perform evaluations on three reference test sets: LOL-v1, LOL-v2-real, and LSRW. Additionally, we assess the performance of model on four non-reference datasets: DICM, LIME, VV, and MEF. We utilize peak signal-to-noise ratio (PSNR), structural similarity index (SSIM) [44] and learned perceptual image patch similarity (LPIPS) [49] for quantitative analysis of reference images. In general, higher values of PSNR and SSIM indicate better image quality. On the other hand, a smaller LPIPS value indicates that the enhanced result is more similar to the reference image. A no-reference image quality evaluator known as the natural image quality evaluator (NIQE) [50] is utilized to assess the quality of non-reference datasets. The lower the NIQE score, the closer the image quality is to a natural high quality image.

Table 2

NIQE scores comparison across the LIME, VV, DICM, MEF datasets. The superior results are highlighted in red, with the second-best achievements in blue.

Methods	LIME	VV	DICM	MEF
MBLLEN [25]	3.512	3.236	3.442	3.366
URetinetex [29]	3.447	2.826	3.339	3.167
SNR [3]	4.942	6.409	3.457	3.337
LLFlow [40]	3.530	2.731	3.421	3.214
HWMNet [11]	3.594	2.355	3.671	3.374
CUE [28]	4.038	2.892	3.418	3.580
Retinexformer [32]	3.925	2.194	3.859	3.251
DiffLL [12]	3.267	3.266	3.014	2.902
EMNet [33]	3.820	3.603	3.625	3.323
PPformer [31]	3.552	2.800	5.754	4.707
BGFlow (ours)	3.240	2.748	3.246	3.168

4.3.1. Quantitative analysis

Table 1 shows the comparative results on four reference datasets. Our BGFlow method achieves significant advantages on several public datasets. Specifically, BGFlow obtains a 0.97 dB PSNR improvement over the second-best method, DiffLL. It reaches a PSNR of 32.23 dB on the LOL-v2-real dataset, which is 2.23 dB higher than the second-best method, EMNet. Our method attains SSIM values of 0.873 on LOL-v1 and 0.920 on LOL-v2-real, surpassing all competitors. Additionally, BGFlow delivers top performance on the Nikon and Huawei datasets, with PSNR values of 17.09 dB and 20.83 dB, respectively. The perceptual quality of our method, as measured by LPIPS scores, also demonstrates its competitiveness, with a score of 0.110 on LOL-v1 and 0.071 on LOL-v2-real, confirming its robustness in enhancing image quality under low-light conditions.

The comparative outcomes across four non-referenced datasets including are summarized in Table 2. Our BGFlow consistently demonstrates superior performance. Specifically, BGFlow demonstrates robust performance on the LIME and DICM datasets, achieving scores of 3.240 and 3.246, respectively. These results highlight its effective improvement of image quality across various lighting conditions. On the VV and MEF datasets, BGFlow achieves a substantial enhancement in natural image quality compared to other techniques. These results reveal that BGFlow has better generalization ability in real scenarios.

Computational complexity. In addition to enhancing visual quality, efficiency is also a crucial factor in image restoration tasks. Table 3 reports the parameter count and average inference time of the compared algorithms. Inference is conducted on the LOL-v1 test set with an image size of 600×400 , and all methods are tested on an NVIDIA RTX 3090Ti GPU. As shown in Table 3, our method achieves a better balance between efficiency and performance, with an average inference time of 0.4047 s, positioning it at a moderate level compared to other methods. Thanks to the proposed wavelet-based flow model, our method reduces the parameter count by 12 times compared to previous flow-based restoration method (LLFlow). We also report the average inference time consumed on the LOL-v1 dataset test set. Notably, our method is 0.2028 s faster per image on average than LLFlow.

¹ <https://sites.google.com/site/vonikakis/datasets>.

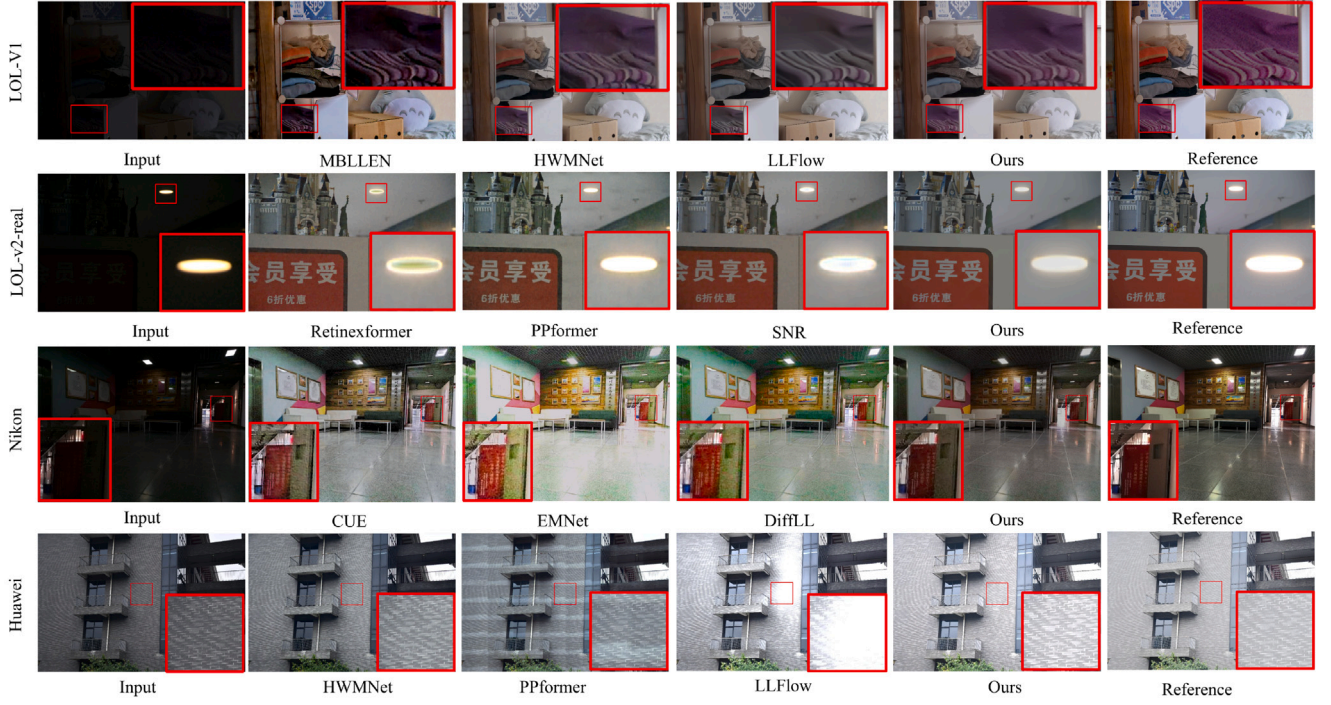


Fig. 6. Comparison of visual results across different methods on four reference datasets (LOL-v1, LOL-v2-real, Nikon, and Huawei datasets in LSRW).



Fig. 7. Comparison of visual results across different methods on four non-reference datasets (DICM, LIME, VV, and MEF datasets).

4.3.2. Qualitative analysis

Figs. 6 and 7 visual comparisons between our method and other methods on reference and non-reference datasets. As can be seen, our results show better contrast, detail accuracy, vivid color, and brightness. Notably, MBLLEN and LLFlow suffer from color distortion on the LOL dataset. Specifically, as can be seen in the cross-dataset

validation visual results in the second to fourth rows of Fig. 6, our BGFLOW effectively mitigates overexposure in bright areas, as well as obtains vivid color with less artifacts. Meanwhile, Retinexformer and SNR introduce some visual artifacts in the brightened regions on the LOL-v2-real dataset. Compared to other methods such as CUE, EMNet, and DiffLL, our method exhibits superior image quality in terms of

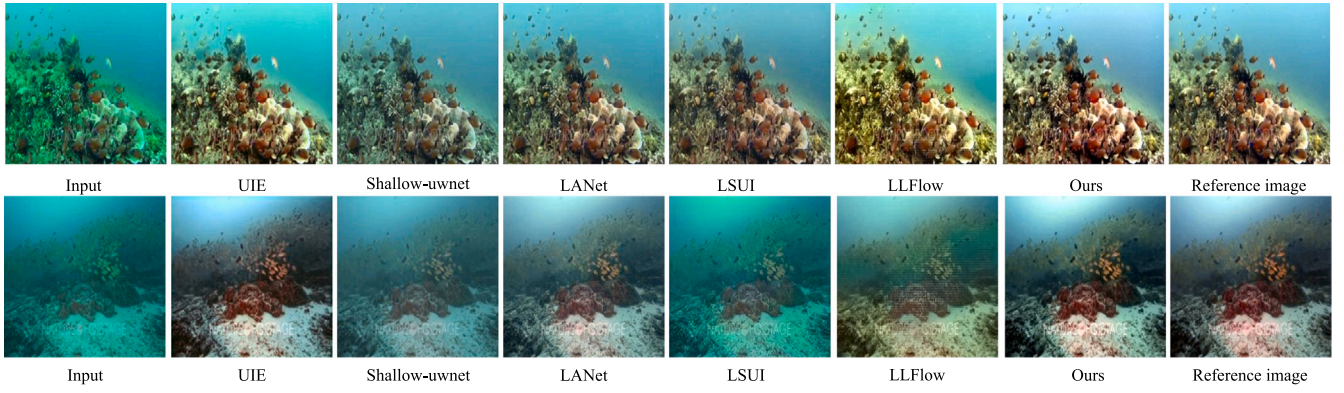


Fig. 8. Visualization results of various approaches on a large-scale underwater image dataset (LSUI). Our method produces competitive visual results. Shallow-uwnet [51] and LANet [52] etc. produce artifacts and color restoration results. Our method accurately restores colors and details, yielding the best results. For the best viewing, please zoom in.

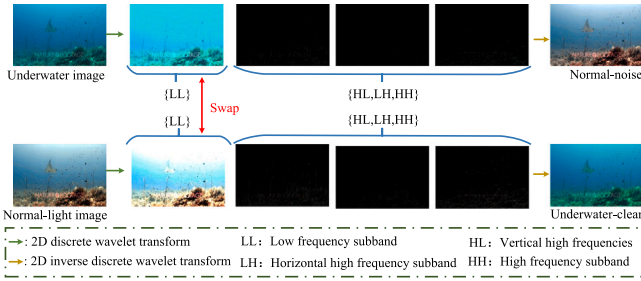


Fig. 9. The characteristics of real underwater images in the wavelet domain. We notice that wavelet transformation can separate brightness and color information. By analyzing the different sub-bands obtained from the wavelet transform, we have found that exchanging the low frequency sub-bands can significantly enhance the brightness and color of underwater images.

Table 3

Comparisons of computational complexity in terms of number of trainable parameters and the average test time on the test images of the LOL-v1 dataset. The best result is highlighted in red, while the second-best result is highlighted in blue.

Methods	Parameters(M)↓	Time (S) ↓
MBLEN [25]	12.09	0.4500
URetinet [29]	0.36	0.0770
SNR [3]	4.01	0.0157
LLFlow [40]	38.86	0.6075
HWMNet [11]	66.56	0.2152
CUE [28]	0.25	0.1348
Retinexformer [32]	1.61	0.0224
EMNet [33]	12.52	0.1521
DiffLL [12]	22.08	0.1837
PPformer [31]	0.10	0.0167
BGFlow (Ours)	3.20	0.4047

color, contrast, and saturation, and our results most closely resemble the reference image. On the Nikon dataset, HWMNet cannot adequately reconstruct the image brightness. PPformer exhibits an unsuccessful enhancement result. In addition, LLFlow over-enhances the exposed area of the image, leading to severe color distortion. Even in regions with complex textures, our output exhibits minimal visual artifacts. Fig. 7 is the visual examples of comparison between our method and other methods on non-reference datasets. As shown in the first row of Fig. 7, our method exhibits more natural enhancement effect in the sky. MBLEN, HWMNet, and LLflow cannot properly reconstruct the sky. As depicted in the second and third rows of Fig. 7, our method achieves more natural visual results. The enhanced results of URetinet and DiffLL exhibit color distortion. As shown in the last row of Fig. 7, EMNet over-enhances the exposed area of the image, making the color of the exposed area seriously distorted. The visual results

Table 4

Application in underwater image enhancement. The full-reference image quality evaluation results on the underwater images from LSUI [53] dataset. The best result is highlighted in red, while the second-best result is highlighted in blue.

Methods	PSNR ↑	SSIM ↑
UIE [54]	20.70	0.837
Shallow-uwnet [51]	17.06	0.765
LANet [52]	19.46	0.820
LSUI [53]	23.68	0.855
LLFlow [40]	21.92	0.827
BGFlow (Ours)	24.52	0.881

demonstrate that our method effectively restores image color while enhancing brightness.

4.4. Underwater image enhancement

In addition to its application in low-light image enhancement tasks, BGFlow can also be leveraged to enhance underwater images. In underwater environments, visibility is typically low. As depth increases, light of different wavelengths is absorbed by the water to varying degrees. Typically, red and orange lights are absorbed first, causing underwater images to appear more blue or green. We analyzed the characteristics of underwater images in the wavelet domain. As shown in Fig. 9, similar to the analysis of low-light images, we exchange the low-frequency sub-bands of the underwater image with those of a normal-light image in the wavelet domain. After the exchange, the underwater image becomes closer to a normal image. This indicates that the low-frequency sub-bands of underwater images typically contain information about overall image brightness, background, and lighting changes. This is similar to how BGFlow enhances low-light images, focusing on extracting brightness characteristics from the low-frequency sub-bands to guide the normalization flow model in reconstructing normal-light image. This is also the reason why our BGFlow is effective for enhancing underwater images.

We conduct a comprehensive evaluation of BGFlow using the underwater dataset LSUI [53]. To ensure a fair experimental comparison, we divide the dataset into 3279 training images and 1000 testing images, retrained all models using publicly available code, and evaluated their effectiveness through both quantitative and qualitative analyses. We qualitatively compare the results of BGFlow with four state-of-the-art underwater image enhancement techniques: UIE [54], Shallow-uwnet [51], LANet [52], LSUI [53]. The superiority of our approach is evident from both the quantitative metrics presented in Table 4 and the visual comparisons depicted in Fig. 8. Our method exhibits excellent performance in various aspects, including color correction, brightness enhancement, and detail restoration.

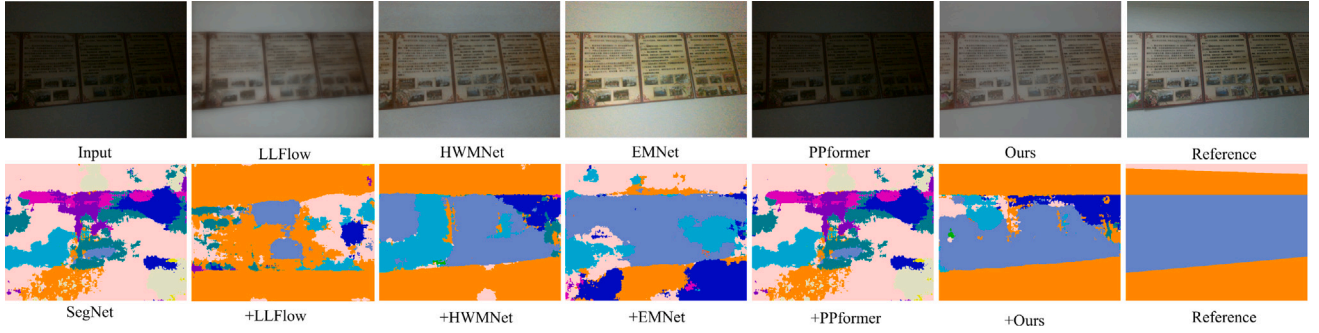


Fig. 10. Comparison of visual results across different methods on LLRGBD-real dataset.

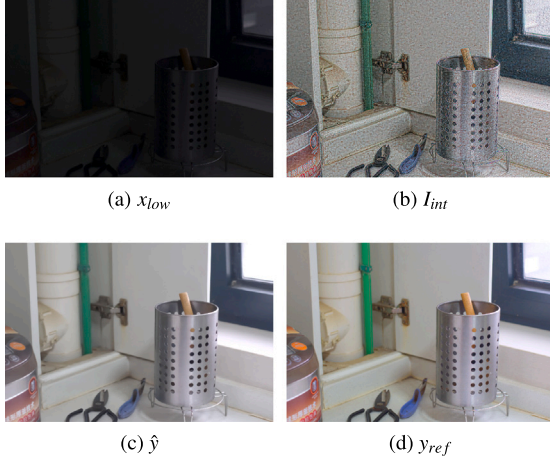


Fig. 11. The visual result of the input image x_{low} , intermediate image I_{int} , output image $\hat{y}$, and reference image y_{ref} .

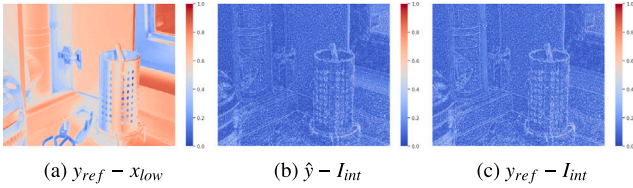


Fig. 12. Heatmap comparing the difference of low-light image to normal-light image, the intermediate image I_{int} to refined result, and I_{int} to normal-light image.

4.5. High-level vision evaluation

To validate the performance of our proposed BGFlow in high-level visual tasks, we select the LLRGBD-real dataset [55], which includes 515 pairs of low-light and normal-light images, with a resolution of 640×480 , captured across 32 indoor scenes (bedrooms, kitchens, bathrooms, living rooms, offices). For the semantic segmentation experiments, we employ SegNet [56] trained on the LLRGBD-real dataset. In our experiments, we process the test data using different low-light image enhancement methods, and then feed these images into SegNet. To evaluate the performance of the proposed BGFlow method in comparison with other approaches, we use three metrics for segmentation assessment: overall accuracy (OA), mean accuracy (mAcc), and mean intersection over union (mIoU). The results shown in Table 5 verify that our BGFlow achieved the best scores. Fig. 10 further demonstrates that SegNet obtained the most accurate segmentation when using BGFlow.

Table 5

Comparison of segmentation accuracy in the low-light images.

Methods	OA	mAcc	mIoU
SegNet [56]	0.292	0.222	0.110
+LLFlow [40]	0.395	0.288	0.167
+HWMNet [11]	0.444	0.327	0.201
+EMNet [33]	0.322	0.256	0.153
+PPformer [31]	0.292	0.222	0.110
+BGFlow (Ours)	0.470	0.365	0.229

Table 6

Ablation studies of different components. The average PSNR/SSIM/LPIPS comparison is conducted on the LOL-v1 datasets, considering effectiveness of proposed blocks. 'w/o' denotes without and 'w/' denotes with.

Methods	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
w/o BGNF	18.57	0.721	0.302
w/o WT	26.71	0.861	0.131
BGFlow	26.84	0.865	0.126

4.6. Ablation study

We perform a comprehensive set of ablation experiments to assess the impact of various components. During the training phase, we strictly follow the implementation details in Section 4.2. Tables 6–8 quantitatively summarize the results of the LOL-v1 dataset in different experimental configurations. A more detailed discussion follows.

Effectiveness of proposed blocks. We conduct ablation experiments to demonstrate the importance of different components in BGFlow. As shown in the first row of Table 6, the enhanced results only using the BFEB. This result reveals the role of the BFEB in the image enhancement task. Furthermore, the effectiveness of the WT in BFEB is verified by replacing the DWT and IDWT with downsampling and upsampling, respectively. The experimental results are presented in the second row of Table 6. The result indicates that WT contributes to improved performance of the model, primarily because it preserves information better than conventional downsampling or upsampling techniques, allowing for more effective multi-scale feature analysis. This capability not only enables more accurate capture of image characteristics but also provides more effective guidance for BGNF.

As shown in Fig. 11, we visualize the output of each network component. Fig. 11(b) shows the intermediate image I_{int} of the BFEB output. It can be seen that the BFEB significantly enhances the brightness and color of the image. This aligns with the original intention of the BFEB design, which is to fully extract the brightness information of the low-frequency subbands to provide conditions for BGNF. Fig. 11(c) illustrates the final output of BGFlow, indicating that the BGNF model can provide further refinement for the intermediate variables. To illustrate the important roles of network components, as shown in Fig. 12, we use a heat map to display the differences across the outputs, succinctly highlighting their respective functions. The three

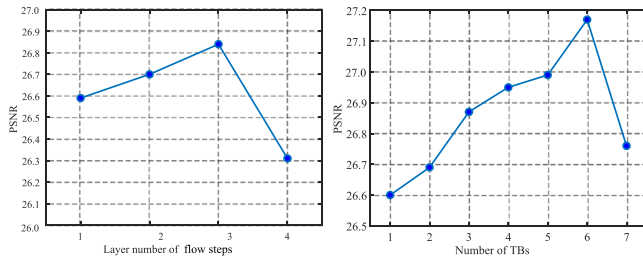


Fig. 13. The enhanced results with respect to the number of flow steps and TBs.

subfigures depict different heatmaps comparing the low-light image with the normal-light image, the intermediate image I_{int} with the refinement result, and the intermediate image I_{int} with the normal-light image. Colors biased towards red indicate more significant differences, whereas colors biased towards blue indicate comparatively smaller differences. Comparing the results in Fig. 12(b) and (c), we observe a striking similarity, indicating that the normalizing flow model accurately simulates the difference between the intermediate image I_{int} and the reference image. This underscores its refinement capabilities.

The impact of varying the number of flow steps and TBs. We discuss the effects of varying the number of flow steps in the normalization flow and the number of TBs in BFEB and the number. Initially, we varied the number of flow steps in each scale of the normalization flow from one to four. The experimental results, illustrated in Fig. 13(a), demonstrate that increasing the number of flow steps further boosts the PSNR value. However, the improvement diminishes gradually and even exhibits performance degradation after a certain number of steps. This indicates that an appropriate number of flow steps can enhance the expressive power of model, while an excessive number of steps increases model complexity, thus affecting its generalization ability. Besides, to achieve a better balance between the number of parameters and performance, in BFEB, TBs ($TB_i, i \in \{1, \dots, I\}$) except TB_1 are all set to 1 and I is empirically set to three, indicating three different scales. This setting is to ensure that TB_1 can better extract rich brightness information and have strong capability to reconstruct the image I_{int} . In detail, we adjusted the number of TBs in TB_1 , ranging from one to seven. Fig. 13(b) depicts the image enhancement performance using the PSNR metric for different configurations. The results indicate that increasing the number of TBs in TB_1 gradually improves the PSNR value. Nevertheless, when the number of TBs exceeds 6, there is a noticeable decline in the PSNR value. This phenomenon may be attributed to the potential issue of overfitting caused by excessively complex model structures. Based on the experiments, we determined the optimal configuration, which entails using 6 TBs in TB_1 and 3 flow steps at each scale in the normalization flow. This configuration enables the model to achieve the best performance in low-light image enhancement tasks while maintaining good computational efficiency and scalability.

The effect of the threshold parameter p . The threshold parameter p in Eq. (8) is a critical hyperparameter that controls the coupling relationship between BFEB and BGNF. Table 7 illustrates the impact of p on the performance of our BGFlow. The experiments show that the best performance is achieved when p is set to 0.2. When p is less than 0.5, I_{int} is selected with a higher probability than y_{ref} , strengthening the coupling between BFEB and BGNF. Conversely, when p exceeds 0.5, I_{int} is selected with a lower probability than y_{ref} , which weakens the coupling relationship between BFEB and BGNF, and leads to a significant decrease in network performance. When p is set to 0, the loss function is designed to minimize the difference between z and I_{int} . Although the coupling relationship between BFEB and BGNF is maintained, the relationship between the outputs of BFEB and BGNF and y_{ref} is neglected, leading to a decline in model performance. In contrast, when p is set to 1, the loss constraint only affects the forward

Table 7

Effect of the threshold parameter p on model performance. The average PSNR/SSIM/LPIPS comparison is conducted on the LOL-v1 dataset, considering the effectiveness of proposed blocks. 'w/o' denotes without and 'w/' denotes with.

Methods	PSNR ↑	SSIM ↑	LPIPS ↓
w/ I_{int} , $p = 0$	23.82	0.780	0.298
w/ I_{int} , $p = 0.2$	26.84	0.865	0.126
w/ I_{int} , $p = 0.5$	24.30	0.822	0.199
w/ I_{int} , $p = 0.8$	18.62	0.561	0.352
w/o I_{int} , $p = 1$	18.67	0.570	0.347

Table 8

The average PSNR/SSIM/LPIPS comparison is conducted on the LOL-v1 datasets, considering the loss function settings. 'w/o' denotes without and 'w/' denotes with.

Methods	PSNR ↑	SSIM ↑	LPIPS ↓
w/o fine-tuning	26.84	0.865	0.126
w/ fine-tuning	27.11	0.873	0.110

process of the BGNF, disregarding the constraint on the I_{int} . At this point, the coupling between BFEB and BGNF does not play a role, resulting in a notable decline in network performance. This confirms the crucial role of I_{int} in BGFlow.

Effectiveness of fine-tuning. The model, which uses a negative log-likelihood function, has achieved significant performance. As illustrated in Table 8, this fine-tuning strategy can result in an increase of 0.18 dB in PSNR and 0.008 in SSIM values. A reduction of 0.016 in the LPIPS score suggests that the perceptual similarity between the two images has improved. The experimental results demonstrate that the fine-tuning method we used can further improve the performance of the model.

5. Conclusion

This paper proposed BGFlow, an innovative brightness-guided normalizing flow model comprising a wavelet-domain encoder and a conditional normalizing flow module. The wavelet-domain encoder has been designed to extract multi-scale brightness features, guiding both training and inference within the flow framework. The brightness feature extraction block has played a crucial role in recovering low-frequency sub-bands from low-light images to generate an intermediate image. BGFlow has ensured similarity between this image and the latent representation, accurately modeling image degradation. The final enhanced result has been seamlessly derived from the latent representation during inference. Quantitative and qualitative evaluations have consistently demonstrated the superiority of BGFlow over state-of-the-art methods, showcasing robust performance in enhancing underwater images. A limitation of the current research is the high computational complexity of the algorithms, which results in low efficiency when processing large-scale data. Future work will focus on optimizing the structure of the algorithms to reduce time complexity and exploring lightweight models.

CRedit authorship contribution statement

Jiale Chen: Conceptualization, Methodology, Software, Validation, Formal analysis, Data curation, Writing – original draft, Writing – review & editing, Visualization. **Qiusheng Lian:** Investigation, Data curation, Writing – review & editing, Validation, Supervision, Funding acquisition. **Baoshun Shi:** Formal analysis, Investigation, Writing – review & editing, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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